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February 3, 1965
Revision of
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January 23, 1958

INTERIM FEDERAL SPECIFICATION

GRINDING MACHINE, BENCH, HAND OPERATED

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20407, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. This specification covers bench-mounted, hand-operated grinding machines used for sharpening edged hand tools.

1.1.1 Federal Specification coverage. Federal Specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and sizes. The grinding machines covered by this specification shall be of the following types and sizes as specified (see tables I and II and 6.2):

Type I. - Heavy duty, with at least 4 bearings.
Size 5. - 5-inch diameter grinding wheel.
Size 6. - 6-inch diameter grinding wheel.
Size 7. - 7-inch diameter grinding wheel.

Type II. - Medium duty, with at least 3 bearings.
Size 4. - 4-inch diameter grinding wheel.
Size 5. - 5-inch diameter grinding wheel.
Size 6. - 6-inch diameter grinding wheel.
Size 7. - 7-inch diameter grinding wheel.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS.

2.1 Specifications and Standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Trible Wall.
PPP-F-320 - Fiberboard; Sheet, Stock and Cut Shapes.

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 224 - Closing, Sealing and Reinforcing of Fiberboard Shipping Containers, General Methods For.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications:

MIL-P-116 - Preservation, Methods Of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Bureau of Standards Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402).

American Standards Association Standards:

B5.17-1958 - Markings for Identifing Grinding Wheels.
B6.1-1932 - Spur Gear Tooth Form.
B6.6-1946 - Gear Tolerances and Inspection.

(Application for copies should be addressed to the American Standards Association, 10 E. 40th St., New York 16, New York).

"Technical soceity and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies."

3. REQUIREMENTS.

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of grinding machines which are otherwise in accordance with the requirements of this specification.

3.2 Preproduction sample. When specified (see 6.2), before production is commenced, a sample of the finished item shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production item.

3.3 Design. The grinding machine shall be designed to operate easily by hand turning of a crank which shall impart rotary motion to a grinding wheel through gearing and shaft. The grinding machine shall be suitable for the purpose intended. All parts subject to wear, breakage, or distortion shall be accessible for replacement, adjustment and repair.

3.3.1 Lubrication. All bearings and mating gears shall be provided with suitable means of lubrication. All oil holes and filler caps shall be easily accessible. Self closing oil caps shall be provided on all external oil holes to prevent grit or dust from entering.

3.3.2 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or machine.

3.3.3 Materiel. Material shall be as specified herein. Material not definitely specified shall be of the quality necessary to produce a machine to meet the requirements specified herein.

3.4 Construction. The machine shall be constructed so that no parts will work loose in service and it shall withstand the requirements specified herein. Construction shall be free from any characteristics or defects that will prevent the machine from passing any of the examinations or tests in Section 4.

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3.4.1 Castings and forgings. All castings and forgings shall be free from defects, scale and mismatching. In no event shall such processes as welding, peening, plugging or filling with solders or metallic pastes be used on castings or forgings for reclaiming any defective parts of the machine.

3.4.2 Welding or brazing. Welding or brazing shall be employed only where those operations are required in the original design. These operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. Such parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.4.4 Surfaces. All surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

3.4.5 Threads. All threaded parts shall be in the inch system and shall conform to National Bureau of Standards Handbook H28.

3.4.6 Gears. All gears and pinions shall be fabricated in the English system and shall conform to American Standards Association Standard ASA B6.6. If tolerances other than those specified are required to manufacture gears or pinions, such tolerances shall assure accuracy of operation of the grinding machine equal to or exceeding that specified herein. Spur gear tooth form shall conform to American Standards Association Standard ASA B6.1.

3.5 Detail of components. The bench grinders shall consist of the following principal component parts:

(a) Screw clamp (see 3.5.1).
(b) Frame (see 3.5.2).
(c) Gears (see 3.5.3).
(d) Shafts (see 3.5.4).
(e) Bearings (see 3.5.5).
(f) Crank and handle (see 3.5.6).
(g) Tool rest (see 3.5.7).
(h) Grinding wheel (see 3.5.8).

3.5.1 Screw clamp. The screw clamp shall be of steel of adequate strength to withstand a force or torsional moment of 120 inch-pounds when applied at right angles to the lonitudinal axis of the screw. The screw shall have a swivel button, at least 1 inch in diameter, which shall adjust itself to the underside of the bench top and shall not revolve as the screw is tightened. The opposite end of the screw shall be fitted securely with a hand knob of irregular shape to facilitate operating and tightening the screw by hand. Screw clamp on type I machines shall accommodate bench tops 1 to 3 inches in thickness. The screw clamp on type II machines shall accommodate bench tops 3/4 to 1-1/2 inches thick.

3.5.2 Frame. The Frame shall be made of cast iron, malleable iron, steel, or aluminum alloy of proper size and strength to support the grinding machine during operation.

3.5.3 Gears. Gears shall have machine-cut teeth. The gears shall be secured to their respective shafts and kept in normal alignment without lateral movement in excess of 1/32 inch when the grinding machine is operated. The gear case for type I machines shall be sealed to prevent loss of lubricant.

3.5.4 Shafts. Shafts shall be of steel. The grinding wheel mounting shaft shall be of sufficient length to accommodate the two wheel-gripping flanges, the grinding wheel and a removable nut with threads to engage mating threads on the shaft. The nut shall fit securely on the threaded part of the shaft. One end of the driving gear shaft shall be shaped for attachment of the crank handle.

3.5.5 Bearings. Bearings shall be of bronze or ball bearings. Bearings shall be installed in proper alignment and center distance so as to allow the gears to operate smoothly and without excessive noise. Sleeve bearings on grinding wheel shaft and hand crank side of driving shaft shall have a length not less than twice the shaft diameter. All sleeve bearings, except lubricant impregnated type, shall have oil grooves and means provided for lubrication.

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3.5.6 Crank and handle. The crank shall be of malleable iron, steel, or aluminum alloy and shall be at least 3-3/4 inches in length between centers of the shaft and the handle spindle holes for sizes 6 and 7 grinders, and at least 3 inches in length for sizes 4 and 5 grinders. The handle shall be of either tough hardwood or metal. The handle shall be attached to the crank by means of a metal spindle. The handle shall rotate freely on the metal spindle. The handle for sizes 6 and 7 grinders shall be not less than 3-1/4 inches long, and the largest diameter not less than 7/8 inch; for sizes 4 and 5 grinders not less than 3 inches long and the largest diameter not less than 3/4 inch. With the drive gears locked to exclude motion of the gear shaft, the crank lever shall withstand a torsional force of at least 150 inch-pounds pressure applied first to the clockwise direction of the crank lever then to the counter colckwise direction of the crank lever. Total slippage, bend or permanent deflection shall not exceed 1/4 inch. A force of 100 inch-pounds applied to the end of the crank in a direction tending to bend the crank away from the gear case shall not break or permanently deform any part of the grinding machine.

3.5.7 Tool rest. The tool rest shall be of malleable iron, steel, or aluminum alloy, and so designed and constructed as to be adjustable to various heights and positions. The tool rest shall be of suitable design for general use in the sharpening of miscellaneous types of edged hand tools.

3.5.8 Grinding wheel. The grinding wheel shall be of aluminum oxide abrasive, vitrified bond. Unless otherwise specified (see 6.2) the grinding wheel shall be medium grain and medium grade of hardness. The wheel shall be square face, straight side type. The wheel shall be mounted on a shaft between two flanged, of sufficient diameter to firmly grip the wheel and prevent slippage during operation. The wheel shall be balanced and properly dressed so as to run true. The grinding wheel shall be marked in accordance with ASA Std. B5.17.

3.6 Performance characteristics. All grinding machines shall operate smoothly and easily without excessive noise or vibration.

3.6.1 Type I, heavy duty grinding machine with four bearings. The type I grinding machine shall be suitable for heavy duty grinding in all classes of work ordinarily performed with tools of this nature. The hand crank shaft and the grinding wheel shaft shall each be provided with a set of two shaft bearings. Each bearing in a set shall be on opposite sides of the gear case. The ratio between the driving and the driven gear shall not be less than 12 to 1 for size 7 grinder, 11 to 1 for size 6 grinder, and 9 to 1 for size 5 grinder. Type I grinders shall be similar to figure 1 and shall conform to the requirements of Table I.

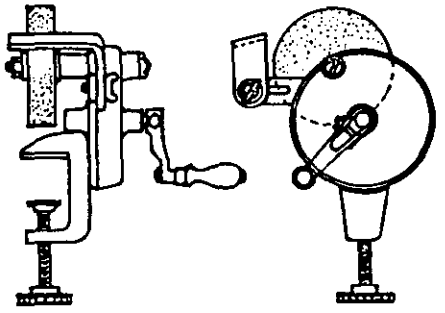


Figure 1. - Type I, heavy duty grinding machine

Table I. - Type I, heavy duty grinding machines.

Size	Grinding wheel			Diameter of shafts (min.)	Net weight of grinding machine (min.)
	Diameter	Thickness	Hole size, minimum		
	Inches	Inches	Inch	Inch	Pounds
5	5	1	3/8	3/8	7
6	6	1-1/4	3/8	3/8	11-1/2
7	7	1-1/2	3/8	3/8	15

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3.6.2 Type II, medium duty grinding machines with three bearings. The type II grinding machine shall be suitable for medium duty grinding in all classes of work ordinarily performed with tools of this nature and shall be provided with a set of bearings on the grinding wheel shaft. The bearings shall be on opposite sides of the gear case. The crank handle shaft on type II machines shall have at least one bearing on the crank handle side of the gear case. (This requirement does not prohibit type I machines with four bearings from acceptance in competition with type II machines). The ratio between the driving and the driven gears shall not be less than 11 to 1 for the sizes 6 and 7 grinders, and not less than 9 to 1 for the sizes 4 and 5 grinders. Type II grinders shall be similar to figure 2 and shall conform to the requirements of table II. When specified (see 6.2) the type II size 6 grinder shall be of such size as to be contained in a tool chest having inside dimensions of 12 by 12 by 12-1/2 inches.

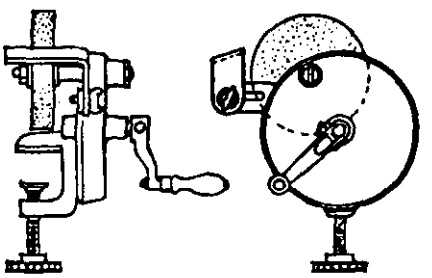


Figure 2. - Type II, medium duty grinder

Table II. - Type II, medium-duty grinding machines.

Size	Diameter	Thickness	Minimum hole size	Diameter of shafts (min.)	Net weight of grinding machine (min.)
	Inches	Inch	Inch	Inch	Pounds
4	4	1	5/16	5/16	3-1/4
5	5	1	3/8	3/8	5
6	6	1	3/8	3/8	7
7	7	1	3/8	3/8	8

3.7 Painting. Unless otherwise specified (see 6.2) the grinding machine shall be painted in accordance with the manufacturer's standard commercial practice.

3.8 Identification marking.

3.8.1 Civil agencies. The grinding machine shall be marked in a plain and permanent manner with the name of the manufacturer or trademark of such known character that the source of manufacture can be readily determined.

3.8.2 Military agencies. The grinding machine shall be marked for identification in accordance with MIL-STD-130.

3.9 Workmanship. Workmanship shall be of the quality necessary to produce a grinding machine of good appearance, having stability and efficient operating characteristics found in first-class commercial items of similar nature.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Preproduction sample inspection. When a preproduction sample is required (see 3.2), inspection and testing of the preproduction sample shall be made of a completely fabricated end item for all provisions of this specification.

4.3 Inspection for acceptance. Inspection shall be made in accordance with the provisions set forth in MIL-STD-105, except where otherwise specified hereinafter. The inspection terms and definitions shall be interpreted in accordance with MIL-STD-109.

4.3.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the specified requirements of referenced subsidiary specification and standards, or, if none, in accordance with this specification.

4.4 Sampling for lot acceptance.

4.4.1 Sampling for inspection. A random sample of grinding machines shall be taken from each inspection lot of material offered for inspection of visual and dimensional characteristics with lot acceptance based on the sampling inspection requirements of table III.

Table III. - Sampling for visual and dimensional inspection of machines AQL (approx.) - 2.5 percent defective

Number of machines in inspection lot	Sample size, number of machines to be inspected (reference MIL-STD-105)
40 and under	Inspection level III
41 to 300	Inspection level II
301 and over	Inspection level I

4.4.2 Sampling for lot acceptance tests. A random sample of grinding machines shall be taken from each inspection lot in accordance with Table IV and shall be subjected to the tests in 4.6.

Table IV. - Sampling for lot acceptance tests.

Number of grinding machines in inspection lot	Number of machines in sample	Acceptance number (defectives)	Rejection number (defectives)
40 and under	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 to 800	25	2	3
801 to 1300	35	3	4
1301 and over	50	4	5

4.5 Inspection.

4.5.1 Visual and dimensional. Each of the sample grinding machines taken in accordance with table III shall be visually and dimensionally inspected to verify compliance with the requirements of this specification. Examination of the grinding machines shall be in accordance with the classification of defects specified in table V. Any grinding machine in the sample containing one or more visual or dimensional defects shall be rejected and if the number of defective grinding machines in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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Table V. - Classification of Defects

Examine	Defect	Major	Minor
Finish	Type of finish not as specified or blistered, pulled, chipped, or area of no film or thin film.		X
	Not finished where required.	X	
	Component or assembly that is painted where serviceability may be affected.		X
Materials	Component not fabricated of the specified material or the type material authorized in the approved sample.	X	
Design	Any characteristic not in accordance with the specified requirements.	X	
Construction and workmanship (general)	Part missing or not specified type and size.	X	
	Fractured, split, punctured, dented, deteriorated, bowed, malformed, or otherwise impaired affecting serviceability	X	
	Defective or otherwise damaged but not affecting serviceability		X
	Misplaced, loose or not in proper alignment affecting serviceability	X	
	Sharp burr, sliver or splinter that may be injurious to personnel or may affect serviceability	X	
	Functioning component that requires abnormal force to operate		X
	Any adjustable assembly that cannot be adjusted or is not properly adjusted to perform the function intended	X	
	Any component not readily accessible for servicing where required		X
Welding and brazing (where required)	Missing, incomplete, burn holes, cracked, fractured or otherwise not fused	X	
	Slag inclusion, slight undercut, not smooth and uniform, scale or flux deposits not removed	X	
Bolts, nuts, screws, studs, and other types of threaded fasteners	Missing, not type specified, broken, stripped, fractured or loose, not affecting serviceability		X
Lubrication (Where applicable)	*Not lubricated where required*	X	
	Insufficently lubricated		X
	Not specified type lubrication fitting, affecting serviceability	X	
	Lubrication fitting not readily accessible for servicing		X
Assembly	Component not properly assembled or secured but not affecting serviceability e.g., Lockwasher missing from metal fastener		X
	Unit perceptibly out of square or alignment	X	
Marking	Missing, incomplete, not legible		X

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4.5.2 Testing. Sample grinding machines selected in accordance with table IV shall be subjected to the test in 4.6. Any machine failing any test shall be rejected and if the number of defective grinding machines in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.6 Test procedures.

4.6.1 Screw clamp test. The grinding machine shall be fastened to a bench top in the usual manner For the purpose of determining whether the clamp is of adequate strength, the screw shall be subjected to a force or torsional moment of 120 inch-pounds under which no parts of the adjusting screw or clamp shall bend, break, or show any visual indications of weakness. In the torsional moment test, a wrench, bar, or clamp may be fastened to the handle of the screw to act as a lever. The longitudinal axis of the arm shall be at right angles to the longitudinal axis of the screw. The force shall be applied through the medium of a spring balance or other load-measuring device, and shall be applied at right angles to the longitudinal axis of the lever. The torsional moment shall be computed by multiplying the force in pounds by the lever length in inches. (Twenty pound-load applied 6 inches from axis of screw = 120 inch-pounds.) The lever length is the distance from the axis of the screw to the place on the lever where the force is applied. In lieu of the test method described, a torque wrench may be used to conduct the test.

4.6.2 Crank test. The grinding machine crank shall be tested as follows: (1) Clamp the driving gear shaft so that it remains stationary. (2) Apply a 150 inch-pound force on the crank handle by using a spring balance, first in the right-hand operating direction of the grinding machine. (3) Mark the position of the crank handle. (4) Apply a 150 inch-pound force on the crank handle in the opposite direction. (5) Measure the distance the crank handle moved in operation "(4)". If the fit of the crank on the shaft, or bending of the crank, permitted a movement of 1/4 inch or greater the sample shall be rejected. (6) A force of 100 inch-pounds shall be applied to the end of the crank in a direction tending to bend the crank away from the gear case. The crank shall not break or show permanent deformation when so tested. The force or torsional moment for the tests shall be computed in a similar manner to that specified in 4.6.1.

4.6.3 Operational test. The grinding machine shall be subjected to a practical grinding test of sufficient duration to satisfy the inspector as to whether the operation is easy and smooth and entirely satisfactory for the purpose intended. There shall be no wheel slippage during grinding operations. Performance of components shall be observed during operation to determine conformance with requirements stated herein.

4.7 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing and marking requirements comply with section 5 of this specification. Defects shall be scored in accordance with table VI. For examination of contents the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be II and the AQL shall be 4.0 defects per hundred units.

Table VI. - Classification of Preparation for Delivery Defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

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5. PREPARATION FOR DELIVERY.

(For civil agency procurement the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102).

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 *Disassembly*. The crank shall be removed from the machine.

5.1.1.2 Preservative application. All exterior unpainted or unplated surfaces subject to corrosion shall be coated with type P-1 preservative in accordance with MIL-P-116.

5.1.1.3 Unit packaging. Each machine shall be packaged in accordance with method IC-5 of MIL-P-116. Fiberboard die-cut blocking and bracing shall be provided with the fiberboard meeting the requirements of PPP-F-320.

5.1.2 Level B. Each machine disassembled in accordance with the usual commercial practice shall be packaged in a close-fitting box conforming to PPP-B-636, class - domestic. Blocking and bracing shall be provided. Closure shall be in accordance with method 170 of Fed. Std. No. 224.

5.1.3 Level C. The machine shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The machines shall be packed in close-fitting boxes conforming to PPP-B-621, class 2, PPP-B-636, V3c, or PPP-B-640, class 2, grade A. The boxes shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight of the triple wall fiberboard box and the wood boxes *shall not exceed* approximately 200 pounds. The gross weight of the PPP-B-636 fiberboard box shall not exceed the weight limitations of the box specification.

5.2.2 Level B. Four unit packages of machines shall be packed in a close-fitting box conforming to PPP-B-636, class - domestic. Closure shall be in accordance with method 170 of Fed. Std. No. 244.

5.2.3 Level C. The machines shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Standard pack (civil agency procurements). The standard pack for civil agency procurements shall be as specified in paragraphs 5.1.2 and 5.2.2 (see 6.3).

5.4 Marking.

5.4.1 Civil agencies. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military activities. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The hand operated bench type grinding machine covered by this specification are intended to be used where it is impractical to provide power driven grinding machines.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2.1).
(c) Preproduction sample, if required (see 3.2).
(d) Grinding wheel grain and grade if other than specified (see 3.5.8).
(e) If the type II, size 6 machine is to be of such size as to be contained in a tool box (see 3.6.2).
(f) Painting if different (see 3.7).
(g) Level of packaging and packing required (see 5.1 and 5.2)

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6.3 Standard pack for civil agencies. The standard pack requirements in 5.3 are intended for use in procurements of stores stock replenishments. Procuring officers should use the standard pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot or intermediate storage point for temporary storage, subsequent issue or shipment to eventual user.

6.4 Transportation description.

6.4.1 Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Grinding machine, bench.

Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Tools, other than power hand tools, set-up.

Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

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